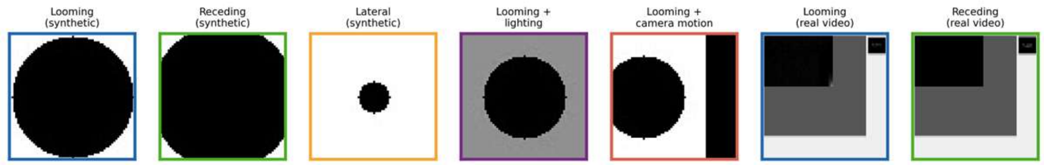
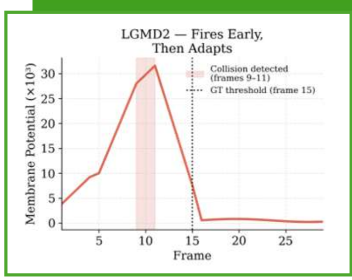
Comparative Evaluation of Biologically Inspired Motion Detection Models: LGMD, EMD, and a Hybrid Spiking Neural Network



LGMD
Lobula Giant Motion Detector

Fires at frames 9–11, before magnitude threshold

- rate-sensitive, fires early, biologically faithful



EMD
Elementary Motion Detector

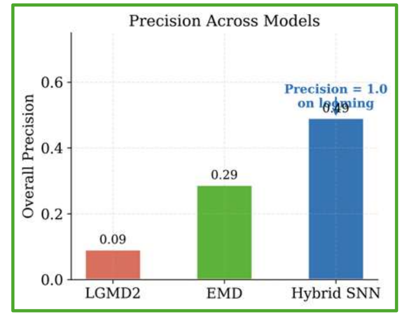
Motion energy ≈ 0 on looming: directional cancellation confirmed

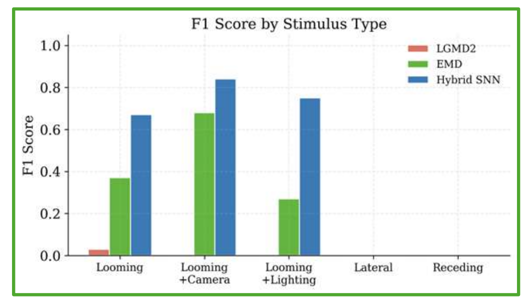
- directional detector, not a collision classifier by design

SNN
Spiking Neural Network

Precision = 1.00 on looming (zero false positives)

- trainable, precision=1.0, sim-to-real gap identified





Hybridising bio-inspired motion detectors within a spiking framework produces reproducible collision prediction with zero false positives on looming stimuli.